

The Sound of Your Secrets

Teaching Your Model to Spy, So You Can Learn to Defend

David vonThenen
AI/ML Engineer



[@davidvonthenen](#)



David vonThenen

- Are you Human or an AI?
- I want 5 Kubernetes
- Virtual Machines are Real
- Cloudy, cloudy, cloudy...
- There is storage for that!

    [@davidvonthenen](https://twitter.com/davidvonthenen)



Agenda

- The Moral Duality Of AI / ML
- Build An Acoustical Keystroke Logger?
 - Dataset Processing Pipeline
 - Model Architecture, Enhancements
 - **Live Demos! Yes, Plural!**
- Real World Attack Vectors
- Privacy, Security, & Protecting Yourself
- Q & A

The Moral Duality of AI / ML

Curing All Disease

► [Pharmaceutics](#). 2024 Oct 14;16(10):1328. doi: [10.3390/pharmaceutics16101328](https://doi.org/10.3390/pharmaceutics16101328) [↗](#)

Artificial Intelligence (AI) Applications in Drug Discovery and Drug Delivery: Revolutionizing Personalized Medicine

[Dolores R Serrano](#)^{1,2,*}, [Francis C Luciano](#)¹, [Brayan J Anaya](#)¹, [Baris Ongoren](#)¹, [Aytug Kara](#)¹, [Gracia Molina](#)¹,
[Bianca I Ramirez](#)¹, [Sergio A Sánchez-Guirales](#)¹, [Jesus A Simon](#)¹, [Greta Tomietto](#)¹, [Chrysi Rapti](#)¹, [Helga K Ruiz](#)¹,
[Satyavati Rawat](#)³, [Dinesh Kumar](#)³, [Aikaterini Lalatsa](#)^{4,5,*}

Editor: Thierry Vandamme

► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

PMCID: PMC11510778 PMID: [39458657](#)

Maximum Money Extraction

ALGORITHMIC
PRICING

where companies hand pr



Instacart

We're excited to announce that we've acquired Eversight, an AI-powered promotions platform for CPG brands and retailers.

This Session...

arXiv > cs > arXiv:2308.01074

Computer Science > Cryptography and Security

[Submitted on 2 Aug 2023]

A Practical Deep Learning–Based Acoustic Side Channel Attack on Keyboards

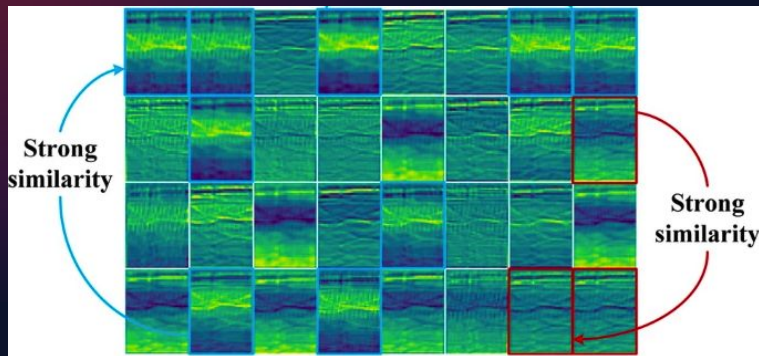
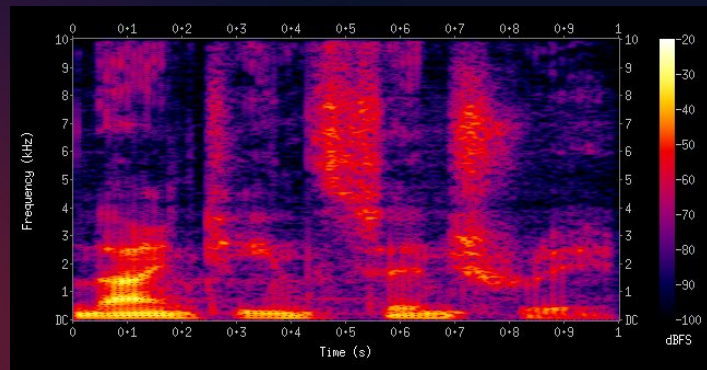
Joshua Harrison, Ehsan Toreini, Maryam Mehrnezhad

With recent developments in deep learning, the ubiquity of micro-phones and the rise in online services via personal devices, acoustic side channel attacks have become a greater threat to keyboards than ever. This paper presents a practical implementation of a state-of-the-art deep learning model in order to detect keystrokes using a smartphone integrated microphone. When trained on keystrokes recorded by a nearby phone, the classifier achieved an accuracy of 95% seen without the use of a language model. When trained on keystrokes recorded using the video-conferencing software Zoom, an accuracy of 85% was achieved, the best for the medium. Our results prove the practicality of these side channel attacks via off-the-shelf equipment and algorithms. We discuss various methods to protect users against these series of attacks.

Building An Acoustical Keystroke Logger

ML Audio Classification?

- MANY Doing This!
- Spectrogram Comparison
 - Visual Representation
 - Similarities Visually
- Example Projects:
 - Cats vs Dogs
 - Environmental Sounds
 - Gunshot Recognition



Always Starts w/ The Dataset...

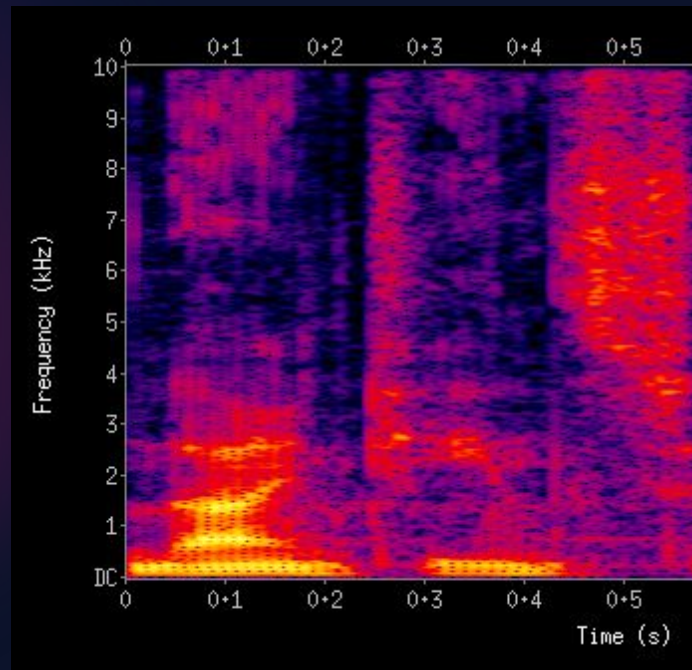
- Keystroke-noiseless-final dataset
 - "Sonotype" challenge at Singapore AI CTF 2024
 - Same problem...
- Multi-Keyboard Acoustic (MKA) Dataset
 - Previous Attempt At This
 - Persistent Inherent Limitations
 - Doesn't Encompass Complexity of Real-World Scenarios
- Will Provide Datasets In GitHub



Data Processing Pipeline

Recordings → Spectrogram Images

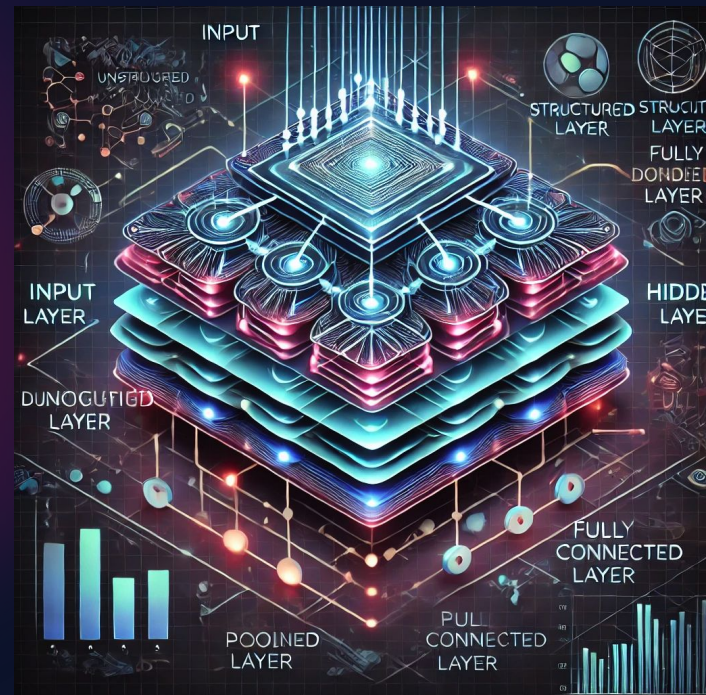
- Directory Name → Classification
- Labels (A-Z, 0-9, Sp, Ent) = 38
- Frequencies Over Time
- RGB (3 channels)
- Resized to a Fixed Grid Data Pipeline
- Normalization (Color + Brightness)
- Robust (Time Shift + Blank Mask)
- Train on "Features" Within The PNG



Model Training

CoAtNet-style Architecture

- Local Patterns, Global Context
- Loss = Cross Entropy
 - Multi-Class Classification
- Uncommon High Epochs (1200)
 - LR $5e-4$, Linear Annealing
 - Optimizer: AdamW (Adam + Weight Decay)
- Compute Train Loss + Train Accuracy Per Epoch



Acoustic Key Detection (Single)

- WAV for Training, Validation, Test Are All The Same!

Train



Validation



Test



Demo: Spectrographic Images

<https://youtu.be/Zn8xEkejHlQ>

Demo: Single Keyboard Train + Inference

<https://youtu.be/iD3GuFPcYN8>

Acoustic Key Detection (Multi)

Lenovo



Keystroke Dependent Dataset

Train

Lenovo



Validation

Lenovo



Test

Lenovo



Demo: Multi Keyboard Train + Inference

<https://youtu.be/1l4XW8JNnww>

Demo: Correcting Probabilities

<https://youtu.be/eXPG-IM3j5M>

Demo: Using AI To Enhance

https://youtu.be/_MG1q0g1-8w

Results From Both Experiments

Single Keyboard

- Accuracy: **100%**



Multiple Keyboards

- Accuracy (Exact): **46%**
- Even Distribution: 3%
 - Random: 1 out of 38

Additional Considerations:

- TopK > 1, Confid. Scores
- All Permutations
- AI to "Fill In The Blanks"

Keystroke Independent Dataset

Train

Lenovo



Validation

Lenovo

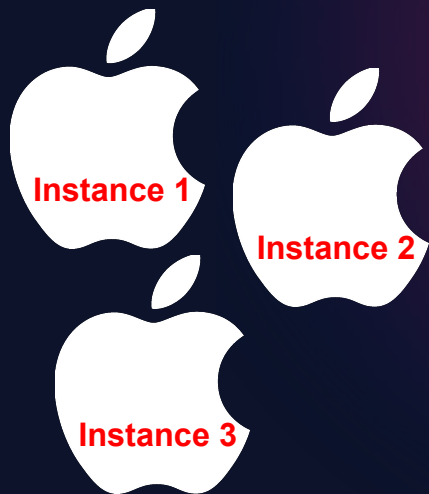


Test

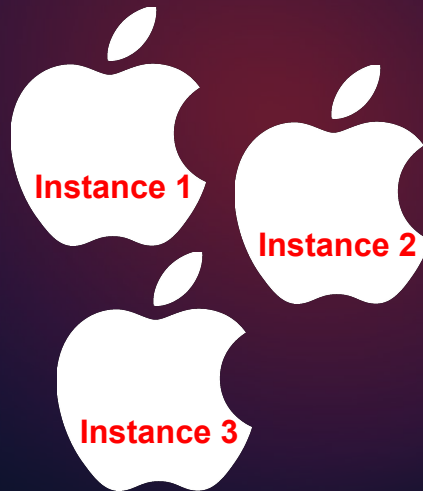


Manufacturing Similarity Dataset

Train



Validation



Test



Letter	Count	Letter	Frequency
E	21912	E	12.02
T	16587	T	9.10
A	14810	A	8.12
O	14003	O	7.68
I	13318	I	7.31
N	12666	N	6.95
S	11450	S	6.28
R	10977	R	6.02
H	10795	H	5.92
D	7874	D	4.32
L	7253	L	3.98
U	5246	U	2.88
C	4943	C	2.71
M	4761	M	2.61
F	4200	F	2.30
Y	3853	Y	2.11
W	3819	W	2.09
G	3693	G	2.03
P	3316	P	1.82
B	2715	B	1.49
V	2019	V	1.11
K	1257	K	0.69
X	315	X	0.17
Q	205	Q	0.11
J	188	J	0.10
Z	128	Z	0.07

Real World Attack Vectors

This Isn't New...



Apple co-founder Steve Wozniak used a toy whistle from Cap'n Crunch cereal to hack AT&T and get free phone calls.

Using Single Keyboard...



What About This...



Credit Card Skimmer



Image Attribution

<https://ktxs.com/news/local/spotting-skimmers-abilene-police-share-tips-to-prevent-debit-and-credit-card-fraud>

Privacy, Security, And Protecting Yourself

2 Factor Authentication



(Strong) Passwords You Don't Know



KeePassXC



1Password



bitwarden

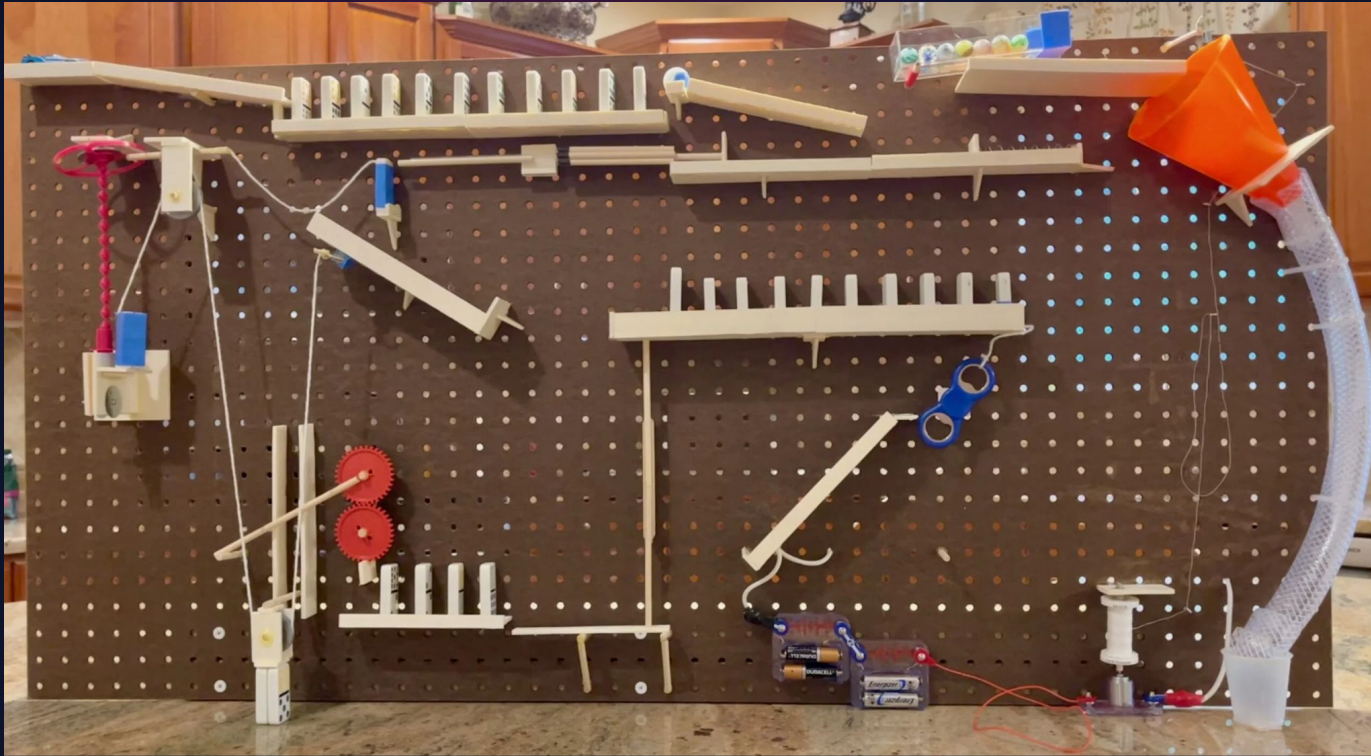
Different Passwords Per Platform



Change Password Frequently



Intentionally Difficult...



Resources

Resources

All Materials (Slides, Code, Instructions, etc):

github.com/davidvonthenen/2026-wearedevelopers-eu-keystroke

Full 50-min Version: [Devovx Greece: Sound of Your Secrets](#)

The Secure Disclosure Podcast - Acoustic Keystroke Loggers:

- [Apple Podcast](#)
- [Spotify Podcast](#)
- [YouTube Podcast](#)

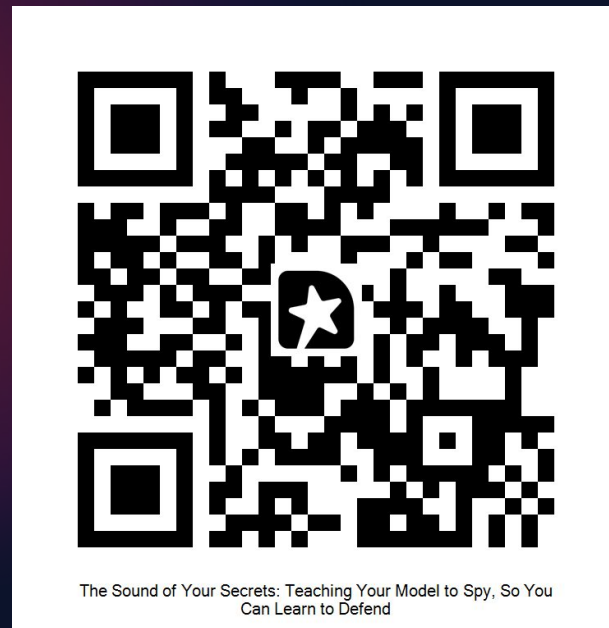
Additional Resources:

- Research: [A Practical Deep Learning-Based Acoustic Side Channel Attack on Keyboards](#)
- Disrupting AI and LLMs: [WeAreDevelopers World Congress 2025: Confuse, Obfuscate, Disrupt](#)



Rate This Workshop!

If You Enjoyed This Session And Found It Informative,
RATE THIS SESSION!



Thank You!



David vonThenen
Senior AI/ML Engineer



@davidvonthenen

